Anthony Allen

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SUMMARY

Automation-focused IT professional with hands-on experience provisioning, configuring, and validating macOS and Ubuntu Server systems for production deployment. Builds Ansible, Python, and Bash automation that cuts manual setup time and catches problems before hardware ships, and uses agentic AI tools (Claude Code) to design and deliver working software quickly. Pursuing a B.S. in Software Engineering at WGU; seeking a DevOps or Systems Engineer role.

EXPERIENCE

T1V, Charlotte, NC
Production Setup and QC October 2025 – Present

- Built a visual regression testing framework for ThinkHub, T1V's large-format touch collaboration software, whose custom rendering engine is invisible to accessibility-based tools like XCUITest and Appium. The suite replays touch and gesture inputs through Peekaboo, captures screenshots at checkpoints, and scores them against baseline images using structural similarity matching, delivering automated testing that the company had pursued for about 10 years.
- Scaled the framework to 35 tests and about 400 checkpoints covering embedded browsers, whiteboards, annotation, multitouch, and session save/restore; one orchestrator runs the full suite and publishes a live dashboard, HTML reports with visual diffs, and CI-style exit codes. Designed and built with Claude Code, including custom skills and subagents that encode how to run the suite, triage failures, and write new tests.
- Modernized the Ubuntu Server relay image: upgraded from Ubuntu 22.04 to 24.04, migrated all remaining Python 2 code to Python 3, and expanded the imaging script to handle most configuration on its own. Retired 5+ years of technical debt and cut hands-on imaging time from about 30 minutes to about 5 minutes per device.
- Built an automated smoke-test pipeline that rebuilds the relay image weekly in a disposable libvirt/KVM virtual machine, validates it against the multisite service, and emails results to the support team, catching breaking changes before they reach customer hardware.
- Removed external dependencies from relay imaging by moving RabbitMQ and Erlang from Homebrew to version-pinned APT packages and bundling an offline package repository, after an upstream update broke production images.
- Deployed Ansible in Docker-based Execution Environments, triggered by a custom Flask phone-home service that automatically registers newly imaged Ubuntu and macOS devices and runs their configuration playbooks.
- Maintain the macOS imaging script across OS releases, including the migration to macOS 27 (TCC permission database schema changes, MAC address filtering). Built a scripted UTM VM build/teardown workflow that resets to a clean test machine almost instantly, compared with about 10 minutes for a bare-metal wipe.
- Developed IntegralConfigurator, a Python/Textual terminal app that auto-detects HDFury Integral 2 devices over USB serial and provides a guided interface for EDID and mode changes; became standard tooling as systems moved from 0–1 to 2–3 Integrals per unit.
- Developed ants-configurator, a Python terminal tool for post-imaging Mac configuration, packaged as a single-file PEX. Making final configurations faster and more accessible for users of all technical levels.
- Image, stage, and QC 20–40 devices per month (Mac Minis, Mac Studios, Ubuntu relay NUCs), verifying hardware and software configuration before customer shipment. Coordinate handoffs with assembly and attend operations and project meetings.
- Troubleshoot hardware and software failures on in-house units and field returns, using system logs and disk diagnostics to find root causes; initiate and track RMAs for faulty components.

Gabe's, Lexington, NC
Retail Supervisor September 2024 – March 2025

- Led and managed store associates, keeping daily operations running smoothly and customers well served.
- Resolved escalated customer issues with fair, practical solutions.
- Troubleshot point-of-sale systems, handheld devices, office computers, and printers, escalating to the company help desk ticketing system when needed.

- Carried out plans to reduce shrink and improve operational efficiency.
- Performed opening and closing duties, including cash office and after-hours alarm response.

Forsyth Technical Community College, Winston-Salem, NC
Web App Developer Intern February 2024 – July 2024

- Built front-end features (HTML, CSS, JavaScript) for a web-based statistical calculator for the math department, working on a team under a senior developer.

HOMELAB & PROJECTS

Self-Hosted Home Lab – *personal environment for learning and experimentation* Ongoing

- Designed and ran a segmented home network: OPNsense firewall with VLANs separating home, development, and isolated/IoT traffic, Unbound DNS, TP-Link Omada wireless managed from a self-hosted controller, and a Twingate connector for remote access without port forwarding.
- Run Proxmox on a retired enterprise desktop hosting about eight VMs and LXC containers, including Kasm Workspaces, Uptime Kuma monitoring, Syncthing, and the Omada controller.
- Use the lab as a DevOps practice platform: hosted a Flask app with Gunicorn behind an Nginx reverse proxy alongside a containerized SQL Server instance, with environment setup automated through Ansible.
- Built a local AI stack: Ollama on an NVIDIA GPU passed through to Docker (nvidia-container-toolkit), serving a VS Code coding assistant and a self-hosted AI agent reachable over Telegram.
- Automated a personal knowledge system with n8n workflows that route Telegram notes and Google Calendar events into Obsidian and generate weekly summaries through the Claude API.

Elevate Retail – *capstone project, Forsyth Technical Community College* January 2025 – May 2025

- Set up development environments (GitHub Codespaces, VS Code Dev Containers) and a containerized MSSQL test database matching production for a 50+ student class building a retail website; documented the builds and troubleshot environment, dependency, and database issues for all teams.
- Built back-end and front-end features with Flask, SQLAlchemy, CSRF protection, Bootstrap 5, Jinja, and AJAX (github.com/Cyber-Kaeh/elevate-retail).

anthonyallen.dev – *personal website* Ongoing

- Designed and deployed a mobile-first site with Vue 3 and Bootstrap 5, with CI/CD through GitHub Actions.

EDUCATION & CERTIFICATIONS

Western Governors University, Online
B.S. Software Engineering Expected June 2027

Forsyth Technical Community College, Winston-Salem, NC
A.A.S. IT – Programming and Software Engineering | High Honors, GPA 4.0 May 2025

CompTIA Project+ July 2026

Microsoft Certified: Azure Data Fundamentals (DP-900) February 2025

TECHNICAL SKILLS

Automation & Config Management: Ansible (Execution Environments, ansible-navigator, ansible-runner), Bash and shell scripting, cron, systemd

Programming: Python (Flask, SQLAlchemy, Textual, Rich), Java, SQL, unit testing, Git, GitHub, GitHub Actions

Operating Systems: Ubuntu Server and Desktop, Debian, Fedora, macOS (imaging, TCC permissions, LaunchDaemons)

Virtualization & Containers: Docker, Dev Containers, libvirt/KVM, UTM, Proxmox, LXC

Networking: VLAN segmentation, OPNsense firewall, DNS (Unbound), DHCP, Nginx reverse proxy, Twingate, TP-Link Omada

Linux Administration: APT package management and version pinning, offline repositories, update management, imaging, SSH, vi/Vim

AI-Assisted Development: Claude Code (agentic workflows, custom skills), Claude API, Ollama, n8n

Collaboration: Jira, HubSpot, Basecamp, Google Workspace, Zoom